

ANALYSIS OF OPEN DATA ENQUIRIES AND COMPLAINTS DATASET USING POWER BI



TASK 01

Authors:

B.R.S.H. Rajamanthri - D/ADC/24/0049

P.D.C.M Jagathchandra D/ADC/24/0032

C.P. Hettiarachchi – D/ADC/24/0009

H.K.T.Uthsarani – D/ADC/24/0001

Date:15/12/2024

Technical Report for Task 1: Analysis of Open Data Enquiries and Complaints Dataset (1 July 2010 – 30 June 2014)

Executive Summary

This technical report aims at offering an analysis of the Open Data Enquiries and Complaints Dataset for the period between 1st July 2010 and 30th June 2014. The records included more than 7000 records that were analyzed systematically using SQL Server for data structuring and Microsoft Power BI for data visualization. Key findings include:

- Out of 1086 complaints, 618 were accepted and 255 were not accepted by the regulatory authorities.
- It is worthy of note that Disability (Physical Disability), Race, and Sexual Harassment were found to be complaint categories.
- It is therefore correct to conclude that outcomes such as conciliation (313 cases) and withdrawal (257 cases) were most common.

This paper explains the process of data cleaning and data transformation, presents the characteristics of the deployed Power BI interactive dashboards, and provides detailed insights. The findings identify areas of systematic problems in the complaint handling process and highlight the directions for improvement for the stakeholders. Further, the report considers the effect of recommendations based on analysis of data for making organizations more responsive and enhancing public satisfaction.



Table of Contents

1. Introduction
2. Objectives
3. Methodology
 - Dataset Overview
 - Data Cleaning and Transformation
 - Tools and Techniques
4. Findings
 - Temporal Trends
 - Complaint Categories and Statuses
 - Resolution Outcomes
5. Dashboard Design and Features
 - Overview
 - Advanced Features
6. Recommendations
7. Conclusion
8. References
9. Appendices

1. Introduction

Complaint management is an important aspect of organizational success since it helps organizations to correct problems, transform processes, and strengthen stakeholder confidence. This paper seeks to analyze certain aspects such as the pattern, trend and the result of the Australian government's "Open Data Enquiries and Complaints Dataset". It covers complaint type, outcome, and the general information of the complainant and covers a period of four years.

Such complaint datasets are usually a mirror of the social, organizational, and policy environment within which they were generated. It is therefore important to spend a lot of time in data preprocessing and choosing the right visualization techniques that will give the insights out of them. As a result, the findings of this report provide stakeholders with knowledge of complaint dynamics and the opportunity to make relevant improvements.

In the analysis, SQL Server is used for preprocessing while Power BI for visualization to make recommendations. This process included organizing raw data and converting it into usable forms in order to analyze wastage and make suggestions for its reduction. The study has implications for policy formulation and implementation, resource mobilization and deployment, and public services improvement. This way, the findings are made understandable and easily usable by both technical and non-technical decision-makers and consumers.

2. Objectives

The primary objectives of this report are:

1. **Identify Trends and Patterns:** Segmenting complaint data into temporal and categorical vectors allows us to reveal fluctuations in complaint frequency, as well as uncover slow-moving trends. By analyzing these trends, we can gain a deeper understanding of the recurring issues faced by the public, identify potential bottlenecks, and pinpoint areas where interventions may be necessary to address persistent problems. This enables organizations to anticipate and proactively respond to emerging challenges, thereby improving overall service delivery.
2. **Visualize Insights:** One of the key goals is to design engaging and intuitive Power BI dashboards that empower users to easily navigate complex datasets. By presenting insights through clear visualizations, users can interact with the data in a way that highlights relevant information and trends. These dashboards are designed to make the data more accessible, ensuring that stakeholders at all levels of the organization can derive meaningful conclusions without needing to delve into technical details. The focus is on enhancing user experience and enabling data-driven decision-making.
3. **Recommend Improvements:** Based on the analysis, the report aims to propose actionable recommendations that can have the most significant impact on improving resolution processes. These include strategies to reduce the number of unresolved cases, streamline complaint management, and ultimately increase public satisfaction. The recommendations will prioritize the adoption of best practices, technological innovations, and workflow optimizations to address the root causes of inefficiencies and enhance the responsiveness of the organization.



Attaining these objectives is essential for extracting meaningful and valuable insights from the raw data, which can be used to guide strategic decision-making within the organization. By providing a structured approach to data analysis and visualization, the report offers a clear roadmap for stakeholders to follow in their efforts to improve service delivery and operational performance. Furthermore, it aims to enhance transparency in the complaint resolution process by providing evidence-based recommendations. This evidence-based approach not only supports organizational goals but also fosters a foundation of trust with the public, reinforcing the organization's commitment to accountability and continuous improvement. Ultimately, this report seeks to equip decision-makers with the insights and tools necessary to create a more responsive, efficient, and trustworthy organizational framework.



3.Methodology

3.1 Dataset Overview

The dataset includes enquiry and complaints data of 7,162 records for four years. Each record includes several key attributes:

- Received Date: The day the complaint was made to the management.
- Method of Submission: Methods like hardcopy, online or email that are available to the complainant and reflecting their choice.
- Complaint Type: Makes a distinction between complaints and inquiries.
- Complaint Status: This field describes the status of the complaint whether it has been accepted, not accepted or the status of the complaint being unresolved.
- Categories: General categories like Disability, Race, Sexual Harassment and others.
- Resolution Outcomes: Explains how a case is resolved through conciliation, withdrawal of any case or referral to another body.

These attributes need to be grasped in order to conduct the segmentation and analysis of data accurately. For instance, sorting out how complaints are distributed by categories makes it easier to allocate the resources and efforts towards solving complaints that matter most.

3.2 Data Cleaning and Transformation

Data cleaning and transformation were crucial in order to make the data more accurate, coherent and appropriate for analysis. Key steps included:

1. **Missing Data Management:** The cases of missing data in fields like “Outcome” and “Category” were handled by logical imputation using patterns that can be seen in other fields of the data. In some instances, records with high missing data values were excluded to ensure validity of analysis was not compromised.
2. **Date Normalization:** Dates were fixed to a common format thus allowing for an effective temporal comparison. Anomalies, for example, when certain months or years, including 2024, had more complaints than others, were identified as candidates for more detailed study with regard to contextual factors.
3. **Categorical Standardization:** Changes in the category names (such as from ‘Disability’ to ‘Physical Disability’) were reconciled to allow for category accumulation and trend analyses.
4. **Derived Metrics:** Other fields were derived, including average of resolution time, monthly complaint volume, and proportion of complaints that are yet to be resolved.
5. **Validation:** The cleaned dataset was checked against the set metrics to confirm accuracy and completeness of the data.

All these steps were well recorded so that other people who wish to replicate the analysis can have a clear view of all the steps taken.

3.3 Tools and Techniques

The analysis employed the following tools and techniques:

1. SQL Server: The SQL queries helped in the selection, sorting as well as aggregation of data.
2. Power BI: Specifically, dashboards were designed for trends and results tracking. Key features included:
 - Summary Pages: Quick analysis based on high-level metrics data.
 - Drill-Through Reports: Category-wise and status wise and regional wise in detail.
 - Custom Visuals: Complex charts for geolocation and time.
3. DAX: Various DAX formulas were applied to calculate year on year complaint trend, resolution rate and other parameters categorized across the complaint types.

4. Findings

4.1 Temporal Trends

The analysis revealed significant temporal variations in complaint submissions:

- **Seasonal Patterns:** Complaints recorded at this level were also found to be time dependent, with some months recording high complaints, due to seasonal factors or organizational events. For example, it may be observed that December and January's high alert levels are due to holidays or organizational inefficiency.
- **Data Anomalies:** The number of complaints received spiked in 2024 and could be indicative of problems with data entry or reporting. Although such cases do not preclude the dataset itself, they indicate the importance of the proper data validation procedures.

4.2 Complaints and Its Status

The distribution of complaints across categories highlighted systemic issues:

Top Categories:

- **Disability (Physical Disability):** 163 complaints (15%).
- **Race:** 134 complaints (12%).
- **Sexual Harassment:** 87 complaints (8%).

Status Analysis:

- **Accepted Complaints:** 589 (57%).
- **Not Accepted Complaints:** 43 (23%).

The review showed that some categories, including Disability and Race, were over-represented in complaints that had not been addressed, suggesting that there may be areas of organizational deafness

4.3 Resolution Outcomes

Resolution outcomes indicated varying degrees of success:

- Conciliation: 313 complaints (29%).
- Withdrawn: 257 complaints (24%).
- Not Initiated: 269 complaints (25%).
- Referred: 134 complaints (12%).

Therefore, there is a need to develop interventions that can address high unresolved rate categories including Disability and Race. Also, withdrawn complaints indicate that complainants may be in a position to be discouraged when seeking redress.

5. Dashboard Design and Features

5.1 Overview

The Power BI dashboards were developed to help users analyze data and make decisions based on it. Key features included:

- **Summary Metrics:** Showed total complaints and complaints by category as well as the resolution of the complaints.
- **Drill-Down Reports:** Allowed the users to sort the complaints based on demography, time, and space.
- **Geospatial Mapping:** The regional disparities of complaints filed and the percentage of their resolutions were also brought into sharper focus.

5.2 Advanced Features

- **Bookmarks:** Thus, predefined paths of navigation helped to reduce the number of actions and increase the relevance of the information for stakeholders.
- **Drill-Through:** Provided rich information on different complaint types and geographies, which allowed for its breakdown.
- **Custom Visuals:** Interpretation improvements using trend lines, clustered bar charts, and heat maps, which helps to consider big data as a wide audience.

6. Recommendations

- **Enhance Data Quality:** Automate the identification of date and category record discrepancies as a way of handling anomalies.
- **Optimize Resolution Processes:** High impact categories that are likely to yield good resolution rates should be targeted, for instance, Disability and Race.
- **Implement Predictive Analytics:** Forecast complaint rates using historical data and mobilize resources for when they are most needed.
- **Increase Public Awareness:** Some of the ways that can be used to withdraw rates include; informing the complainants about the available paths to the resolution of the complaint.
- **Target Regional Disparities:** This is because the regional areas that receive many complaints need more resources to be assigned to them.

7. Conclusion

The paper shows that proper data preprocessing and the choice of appropriate visualization methods play the essential role in extracting valuable information from large and frequently, heterogeneous data sets. In the present study, by elaborating these techniques, we have explained how such techniques help to interpret the raw data to create valuable information. These tactics in data cleaning, transformation, and visualization enable stakeholders to obtain adequate understanding of emerging trends, as well as the general patterns inherent in organizational performance decision-making.

Applying the solutions that can be generated from the analysis described in this paper will help the organizations improve the current complaint handling mechanisms that are in place to serve the public. Besides increasing organizational effectiveness this approach also serves to enhance the perception that the organization has towards the public and by extension enhance confidence in the organizations capacity to reply and deal with issues in the shortest time. By implementing these strategies, stakeholders can build sustainable best practice to improve service delivery consequently, increase public satisfaction and affirm the stakeholders' confidence in the organs assertion of responsibilities. In the long run, this process will not only make the organization to become efficient but also create a better and more effective relationship with the community it serves.

8. References

1. Open Data Enquiries and Complaints Dataset. (Ref: <https://data.gov.au/dataset/ds-sa-aff12383-bb28-4a97-9c75-fbbcf9eeec7e/details?q=gender>)
2. Guidelines for Technical Report Writing, Institution of Engineering and Technology (IET). (Ref: <https://www.theiet.org/media/5182/technical-report-writing.pdf>)

9. Appendices

Appendix A: SQL Queries

```
Assignment 03 Task...08F131P\USER (59) X
SELECT *
FROM DiscriminationComplaints.dbo.opendataeocenquiriesandcomplain$;

--Checking for NULLs
SELECT *
FROM DiscriminationComplaints.dbo.opendataeocenquiriesandcomplain$
WHERE
    [Received date] IS NULL OR
    [How matter received] IS NULL OR
    [Matter Type] IS NULL OR
    [Complaint status] IS NULL OR
    [Grounds] IS NULL OR
    [Areas] IS NULL OR
    [Events] IS NULL OR
    [Complainant / Enquirer Gender] IS NULL OR
    [Complainant / Enquirer Birth Country] IS NULL OR
    [Complainant / Enquirer ATSI] IS NULL OR
    [Complainant / Enquirer Age Group] IS NULL OR
    [Interpreter required] IS NULL OR
    [Incident Industry Type] IS NULL OR
    [How found out about EOC] IS NULL OR
    [Matter Outcome] IS NULL OR
    [Outcome Reasons (complaints only)] IS NULL;

--Replacing missing values
SELECT TOP (1000)
```

110 %

Connected. (1/1) DESKTOP-08F131P\MSSQLSERVER... DESKTOP-08F131P\USER (59) master 00:00:00 0 rows

```
Assignment 03 Task...08F131P\USER (59) X
[Matter Outcome] IS NULL OR
[Outcome Reasons (complaints only)] IS NULL;

--Replacing missing values
SELECT TOP (1000)
    ISNULL([Received date], '1900-01-01') AS [Received date],
    ISNULL([How matter received], 'Unknown') AS [How matter received],
    ISNULL([Matter Type], 'Unknown') AS [Matter Type],
    ISNULL([Complaint status], 'Unknown') AS [Complaint status],
    ISNULL([Grounds], 'Unknown') AS [Grounds],
    ISNULL([Areas], 'Unknown') AS [Areas],
    ISNULL([Events], 'Unknown') AS [Events],
    ISNULL([Complainant / Enquirer Gender], 'Unknown') AS [Complainant / Enquirer Gender],
    ISNULL([Complainant / Enquirer Birth Country], 'Unknown') AS [Complainant / Enquirer Birth Country],
    ISNULL([Complainant / Enquirer ATSI], 'Not Specified') AS [Complainant / Enquirer ATSI],
    ISNULL([Complainant / Enquirer Age Group], 'Not Specified') AS [Complainant / Enquirer Age Group],
    ISNULL([Interpreter required], 'No') AS [Interpreter required],
    ISNULL([Incident Industry Type], 'Unknown') AS [Incident Industry Type],
    ISNULL([How found out about EOC], 'Unknown') AS [How found out about EOC],
    ISNULL([Matter Outcome], 'Pending') AS [Matter Outcome],
    ISNULL([Outcome Reasons (complaints only)], 'Not Specified') AS [Outcome Reasons (complaints only)]
FROM DiscriminationComplaints.dbo.opendataeocenquiriesandcomplain$;
```

110 %

Connected. (1/1) DESKTOP-08F131P\MSSQLSERVER... DESKTOP-08F131P\USER (59) master 00:00:00 0 rows

Appendix B: Dashboard



